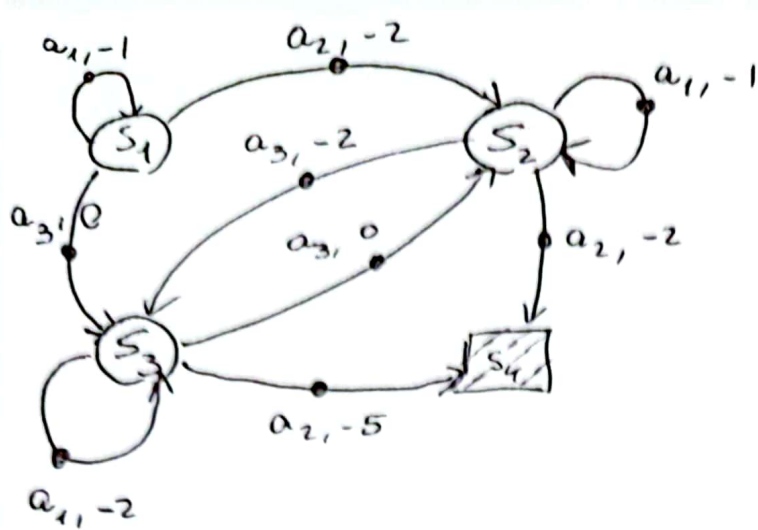




$$\boxed{2}$$

$$\pi_*(a_1, s_j) = 0$$

$$\pi(a_2, s_j) = \pi(a_3, s_j) = \frac{1}{2}$$

$$V(s_1) \leftarrow \frac{1}{2} \left(\overset{a_2}{-2} + V(s_2) \right) + \frac{1}{2} \left(\overset{a_3}{0} + V(s_3) \right) = \frac{1}{2} (-2 + V(s_2) + V(s_3))$$

$$V(s_2) \leftarrow \frac{1}{2} \left(\overset{a_2}{-2} + V(s_4) \right) + \frac{1}{2} \left(\overset{a_3}{-2} + V(s_3) \right) = \frac{1}{2} (-4 + V(s_3))$$

$$V(s_3) \leftarrow \frac{1}{2} \left(\overset{a_2}{-5} + V(s_4) \right) + \frac{1}{2} \left(\overset{a_3}{0} + V(s_2) \right) = \frac{1}{2} (-5 + V(s_2))$$

$$V = (0, 0, 0, 0)$$

$$\boxed{k=1}$$

$$V(s_1) \leftarrow \frac{1}{2} (-2) = -1$$

$$V(s_2) \leftarrow \frac{1}{2} (-4) = -2$$

$$V(s_3) \leftarrow \frac{1}{2} (-5 - 2) = -\frac{7}{2}$$

$$V = (-1, 0, 0, 0)$$

$$V = (-1, -2, 0, 0)$$

$$V = (-1, -2, -\frac{7}{2}, 0)$$

$$\boxed{k=2}$$

$$V(s_1) \leftarrow \frac{1}{2} \left(-2 - 2 - \frac{7}{2} \right) = -\frac{15}{4}$$

$$V(s_2) \leftarrow \frac{1}{2} \left(-4 - \frac{7}{2} \right) = -\frac{15}{4}$$

$$V(s_3) \leftarrow \frac{1}{2} \left(-5 - \frac{15}{4} \right) = -\frac{35}{8}$$

$$V = \left(-\frac{15}{4}, -2, -\frac{7}{2}, 0 \right)$$

$$V = \left(-\frac{15}{4}, -\frac{15}{4}, -\frac{7}{2}, 0 \right)$$

$$V = \left(-\frac{15}{4}, -\frac{15}{4}, -\frac{35}{8}, 0 \right)$$

$$\boxed{k=3}$$

$$V(s_1) \leftarrow \frac{1}{2} \left(-2 + \frac{15}{4} - \frac{35}{8} \right) = \frac{1}{2} \frac{-16-30-35}{8} = -\frac{81}{16}$$

$$V = \left(-\frac{81}{16}, -\frac{15}{4}, -\frac{35}{8}, 0 \right)$$

$$V(s_2) \leftarrow \frac{1}{2} \left(-4 - \frac{35}{8} \right) = -\frac{67}{16}$$

$$V = \left(-\frac{81}{16}, -\frac{67}{16}, -\frac{35}{8}, 0 \right)$$

$$V(s_3) \leftarrow \frac{1}{2} \left(-5 - \frac{67}{16} \right) = -\frac{147}{32}$$

$$V = \left(-\frac{81}{16}, -\frac{67}{16}, -\frac{147}{32}, 0 \right)$$

$$\boxed{4} \quad V_{\pi} = \left(-\frac{11}{2}, -\frac{13}{3}, -\frac{14}{3}, 0 \right)$$

$$\pi'(s_1) = \operatorname{argmax}_a \begin{bmatrix} a_1: -1 + V(s_1) \\ a_2: -2 + V(s_2) \\ a_3: a + V(s_3) \end{bmatrix} = \operatorname{argmax}_a \begin{bmatrix} a_1: -13/2 \\ a_2: -19/3 \\ a_3: -14/3 \end{bmatrix} = a_3$$

$$\pi'(s_2) = \operatorname{argmax}_a \begin{bmatrix} a_1: -1 + V(s_2) \\ a_2: -2 + V(s_2) \\ a_3: -2 + V(s_3) \end{bmatrix} = \operatorname{argmax}_a \begin{bmatrix} a_1: -16/3 \\ a_2: -2 \\ a_3: -20/3 \end{bmatrix} = a_2$$

$$\pi'(s_3) = \operatorname{argmax}_a \begin{bmatrix} a_1: -2 + V(s_3) \\ a_2: -5 + V(s_3) \\ a_3: 0 + V(s_2) \end{bmatrix} = \operatorname{argmax}_a \begin{bmatrix} a_1: -20/3 \\ a_2: -5 \\ a_3: -13/3 \end{bmatrix} = a_3$$

$$[5] \quad V_Q = \left(-\frac{15}{2}, -\frac{17}{3}, -\frac{19}{3}\right)$$

$$\theta'(s_1) = \underset{a}{\operatorname{argmax}} \begin{bmatrix} a_1: -1 + V(s_1) \\ a_2: -2 + V(s_2) \\ a_3: 0 + V(s_3) \end{bmatrix} = \underset{a}{\operatorname{argmax}} \begin{bmatrix} a_1: -19/2 \\ a_2: -23/3 \\ a_3: -19/3 \end{bmatrix} = a_3$$

$$\theta'(s_2) = \underset{a}{\operatorname{argmax}} \begin{bmatrix} a_1: -1 + V(s_2) \\ a_2: -2 + V(s_4) \\ a_3: -2 + V(s_3) \end{bmatrix} = \underset{a}{\operatorname{argmax}} \begin{bmatrix} a_1: -20/3 \\ a_2: -2 \\ a_3: -25/3 \end{bmatrix} = a_2$$

$$\theta'(s_3) = \underset{a}{\operatorname{argmax}} \begin{bmatrix} a_1: -2 + V(s_3) \\ a_2: -5 + V(s_4) \\ a_3: 0 + V(s_2) \end{bmatrix} = \underset{a}{\operatorname{argmax}} \begin{bmatrix} a_1: -25/3 \\ a_2: -5 \\ a_3: -17/3 \end{bmatrix} = a_2$$

$$\theta' \neq \pi'$$